

Introduction

The Kjeldahl nitrogen analysis is a well established technique and has become the standard method for the quantitative determination of organic-bound nitrogen (Classical Kjeldahl Analysis, **CKA**).

The **CKA** technique is the *standard procedure* for protein determination in food (Literature: 10). The analysis is based on three steps:

1) Kjeldahl digestion: The sample is strongly heated in a Kjeldahl digestion flask containing 98% sulfuric acid, H_2SO_4 , during 2-3 hours. Nitrogen is released from organic material and is present in the digested solution as ammonium sulfate, $(\text{NH}_4)_2\text{SO}_4$.

2) Distillation of ammonia: To determine the nitrogen content in the digested solution, concentrated sodium hydroxide solution (30% NaOH) is added. Ammonia (NH_3) is generated according to:



Ammonia is distilled using water steam and collected in a 1-2% boric acid solution, H_3BO_3 :



3) Titration: Ammonium tetraborate, $(\text{NH}_4)_2\text{B}_4\text{O}_7$, can be titrated with hydrochloric acid (HCl):



Literature: 1, 2, 3, 4, and 5.

The ammonia distillation step represents an additional source of error and work and leads also to an increase of the analysis time. This step can be avoided by **direct determination** of nitrogen using the two following alternative determinations:

- Ammonia determination by ISE (**NH₃-ISE**):** NaOH reacts with ammonium sulfate in the digested solution to generate ammonia (NH_3), sodium sulfate (Na_2SO_4) and water (see step 2). Thus, ammonia can be directly determined by using an ammonia gas-sensing electrode (NH_3 -ISE). Ammonia molecules diffuse through the membrane of the NH_3 -ISE leading to a change of the pH-value of the inner electrolyte. The change is detected by a glass pH-electrode mounted inside the ammonia selective electrode.
(Literature: see ref. 6 and 4)
- Urotropine reaction (**HMTA- Method**):** Nitrogen is determined by first reacting ammonium with 37% formaldehyde to give urotropine ($(\text{CH}_2)_6\text{N}_4$, Hexamethylene-tetramine, HMTA) in the digested solution:
$$2 (\text{NH}_4)_2\text{SO}_4 + 6 \text{HCHO} \rightleftharpoons (\text{CH}_2)_6\text{N}_4 + 2 \text{H}_2\text{SO}_4 + 6 \text{H}_2\text{O}$$

The generated sulfuric acid is determined by titration with NaOH.
(Literature: see ref. 7, 8, 9 and 4).

In this application brochure, the classical Kjeldahl analysis (**CKA**), the direct ammonia determination (**NH₃-ISE**), and the urotropine reaction (**HMTA**) have been compared with respect to **1) accuracy** (expressed as recovery, $\bar{x}$) and **2) reproducibility** (rel. standard deviation, RSD).

Overview of the techniques

- | | |
|------------------------------|--|
| A) CKA | 1) Kjeldahl digestion
2) Ammonia distillation with water steam
3) Titration with hydrochloric acid
<i>simplified scheme:</i> $(\text{NH}_4)_3\text{BO}_3 + 3 \text{HCl} \rightleftharpoons 3 \text{NH}_4\text{Cl} + \text{H}_3\text{BO}_3$ |
| B) NH₃-ISE | 1) Kjeldahl digestion
2) Direct determination by NH ₃ -ISE |
| C) HMTA-Method | 1) Kjeldahl digestion
2) Direct titration of ammonium
This determination consists of 4 steps:
- Neutralization to pH 3.2
- Titration to the equivalence point
- Addition of 6 ml 37% formaldehyde previously neutralized
- Titration with NaOH to neutralize the produced sulfuric acid. |

Based on these three analytical procedures, titration methods have been developed to perform automated nitrogen determination. Priority was given to *the accuracy* and *the precision* achievable rather than to a *fast analysis procedure*. If necessary, the analysis time can be shortened by modifying the corresponding control parameters.

First, the methods were tested using ammonium sulfate in sulfuric acid. Then the methods were compared by determining the nitrogen and protein content in different dairy products (milk, cheese, yoghurt...). The protein content was calculated by multiplying the nitrogen content by a factor **f** which depends on the sample, e.g., **f = 6.38** for dairy products.

Applications for nitrogen determination by Kjeldahl digestion

- | | | | |
|------------------------------|---------------------------------|-------------|--|
| A) CKA | 3) Titration with HCl | M120 | Test with $(\text{NH}_4)_2\text{SO}_4$ |
| | | M121 | Dairy products |
| B) NH₃-ISE | 2) Direct measurement | M122 | Test with $(\text{NH}_4)_2\text{SO}_4$ |
| | | M123 | Dairy products |
| | | M124 | NH ₃ -ISE Calibration |
| C) HMTA-Method | 2) Direct titration of ammonium | M125 | Test with $(\text{NH}_4)_2\text{SO}_4$ |
| | | M126 | Dairy products |